CFRS-DA:End-to-End Domain Adaptation Semantic Segmentation Framework for Cloudy Remote Sensing

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***Abstract*—Semantic segmentation has emerged as a prominent research direction due to its pixel-level precision and scene parsing capabilities. Satellite remote sensing images often suffer from significant cloud occlusion, posing challenges in label annotation for semantic segmentation. Current methodologies typically adopt a two-stage approach involving separate cloud removal preprocessing followed by segmentation, which lacks end-to-end solutions and compromises efficiency. This paper proposes an end-to-end framework for cloud-occluded satellite image analysis: the Adversarial Entropy Minimization with Curriculum Self-Training Domain Adaptation Framework (AEM-CurST-DA). The framework first introduces an Atrous Convolution Pyramid Filtering Feature Fusion Encoder Network (ACPF²Net), which suppresses cloud interference through dilated convolutions and multi-level feature fusion, eliminating conventional preprocessing. Subsequently, an entropy adversarial domain discriminator and curriculum self-training strategy are employed to guide cross-domain feature alignment by minimizing inter-domain confidence entropy discrepancies and selecting high-quality pseudo-labels through entropy filtering. Hybrid source-target domain training further mitigates domain shift. Experimental validation achieves over 50% mIoU on cross-domain Potsdam and Vaihingen datasets, with mAP scores of 88.3% (Level-1) and 74.3% (Level-2) on a custom marine vessel dataset, demonstrating the framework's effectiveness in achieving efficient semantic segmentation under cloud-occluded conditions.**

Keywords-Remote Sensing; Unsupervised Domain Adaptation; Semantic Segmentation; Cloud/fog conditions

I. INTRODUCTION

Maritime ship detection, classification, and localization via optical remote sensing imagery are critical for integrated space-air-sea information systems, demanding higher precision than general targets. However, pervasive cloud interference severely limits performance.

Current dehazing methods primarily estimate atmospheric light and transmission in degradation models. Zhang et al.[1] proposed DCPDN with dual subnetworks, while Li et al.[2] reformulated the model for single-parameter estimation (AOD-Net). Qin et al.[3] focused on predicting haze-free images or residuals. Though effective, these methods often introduce distortion during restoration. For cloud removal, Song et al.[4] adapted DeHazeFormer with modified normalization and spatial aggregation in Transformers, yet distortions persist.In domain-adaptive semantic segmentation. Mazza et al.[5] employed fine-tuning for small domain gaps, while self-training faces challenges in pseudo-label quality and utilization. Current pseudo-label acquisition in unsupervised domain adaptation primarily relies on the direct generation paradigm employed in CBST[6] and IAST[7], where confidence-based thresholding mechanisms dominate the label prediction process. Current approaches include entropy minimization (Wang et al.[8], Vu et al.[9], Zhang et al.[10]), generalized adaptation (Qu et al.[11], Chen et al.[12], Peng et al.[13]), and multi-task frameworks (MFANet[14], MuLT[15]). However, these methods, designed for large-scale ground images, underperform on remote sensing data due to limited sample sizes and distinct feature extraction needs.

However，prior research predominantly focused on clear images or ideal weather conditions, demonstrating poor adaptability to complex meteorological scenarios in remote sensing. Although image restoration techniques (e.g., dehazing/cloud removal) have been explored, they inevitably introduce distortions during recovery, where information loss hinders subsequent image understanding. Furthermore, existing approaches typically isolate dehazing and semantic segmentation as independent tasks, concentrating on one aspect while neglecting the critical role of end-to-end networks under such conditions. To address annotation scarcity in remote sensing and meet precise maritime ship localization requirements, this work proposes a domain-adaptive semantic segmentation framework. Our algorithm enables cross-domain execution for ship detection, classification, and localization in cloud-obscured maritime imagery without distortion-prone restoration.

II. METHON

A. Domain Adaptation Framework AEM-CurST-DA

To address the lack of annotated data for cloud/fog-affected remote sensing imagery, we design an unsupervised domain adaptation framework (AEM-CurST-DA, Fig.1) for end-to-end cross-domain semantic segmentation under atmospheric interference. The framework integrates three components: a

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curriculum self-training strategy, an entropy-aware adversarial domain discriminator, and the ACPF²Net model.

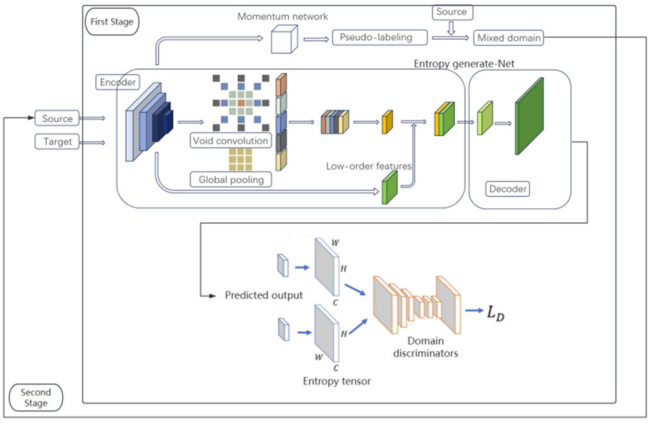


Fig 1 Domain Adaptation Framework AEM-CurST-DA

The framework integrates self-training with curriculum learning principles, implementing unsupervised domain adaptation (UDA) through two phases bridged by an intermediate domain. Specifically, a momentum network generates pseudo-labels, which undergo entropy-based filtering to ensure high quality. Unlike conventional curriculum learning, our key innovation lies in blending filtered pseudo-labels with source domain data during the challenging phase to form hybrid training sets. This design offers dual advantages:

Quality Control: Entropy-aware pseudo-label selection prevents low-confidence predictions from corrupting the training process.

Distribution Balance: Hybridization maintains sufficient training samples while aligning data distributions, effectively mitigating class-imbalance issues that degrade rare-class recognition.

B. Confidence Entropy and Domain Discriminator

In the AEM-CurST-DA framework, confidence entropy (Eq. 2) is introduced as an evaluation metric to filter high-quality pseudo-labels for hybrid training with the source domain, while also serving as input to the domain discriminator.

Let X_s denote the source domain image dataset and Y_s represent its pixel-wise annotations. For an input image $x_s \in X_s$ with label $y_s \in Y_s$, the network F generates a C-dimensional pixel-level probability vector $\mathbf{P}_x^{(h,w,c)}$ after the softmax layer, representing class probability distributions. The segmentation loss is formulated as:

$$L_{seg}(x_s, y_s) = \sum\sum\sum y_s^{(h,w,c)} \log P_{x_s}^{(h,w,c)} \quad (1)$$

For the target domain lacking annotated samples Y_t, direct supervised training becomes infeasible. While existing methods utilize model predictions as pseudo-labels by selecting sufficiently confident pixels, our approach further constrains the model to inherently generate high-confidence predictions through entropy minimization. Specifically, we introduce a Shannon entropy loss L_{ent} to maximize prediction certainty. Given a target domain image xt, the entropy map $Ex_t \in [0,1]^{H \times W}$ at pixel (h,w) is computed as:

$$E_{x_t} = \frac{-1}{log(C)} \sum_{1}^{C} P_{x_t}^{(h,w,c)} \log P_{x_t}^{(h,w,c)} \quad (2)$$

The entropy loss L_{ent} is defined as the summation of pixel-wise normalized entropy across all spatial positions:

$$L_{ent}(x_t) = \sum E_{x_t} \quad (3)$$

The joint network parameter optimization with entropy regularization is formulated as the following minimization objective during training:

$$min \frac{1}{|X_s|} \sum_{x_s} L_{seg}(x_s, y_s) + \frac{\lambda ent}{|X_t|} \sum_{x_t} L_{ent}(x_t) \quad (4)$$

To mitigate domain shift between source and target domains through entropy-driven optimization, the input formulation for the generative component is defined as:

$$I_x = -P_x \cdot log P_x \quad (5)$$

Integrating the domain adversarial learning paradigm from generative adversarial networks (GANs), our framework employs the discriminator in Fig. 2 to achieve indirect entropy minimization through adversarial training. Let L_D denote the cross-entropy domain classification loss. The discriminator's training objective is formulated as:

$$min \frac{1}{|X_s|} \sum_{X_s} L_D(I_{X_s}, 1) + \frac{1}{|X_t|} \sum_{X_t} L_D(I_{X_t}, 0) \quad (6)$$

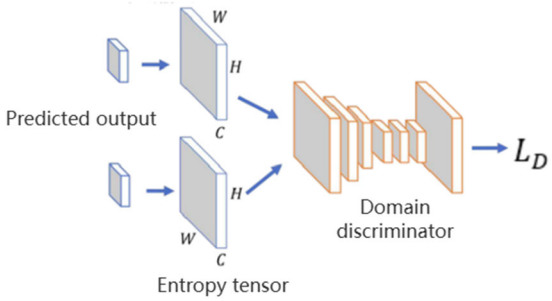


Fig 2 Domain Discriminator

Consequently, the adversarial objective for training the segmentation network is formulated as:

$$min \frac{1}{|X_t|} \sum_{X_t} L_D(I_{X_t}, 1) \quad (7)$$

C. Feature Fusion Encoder Network Model ACPF²Net

While convolutional neural networks (CNNs), particularly ResNet variants, efficiently capture local details and positional cues, they struggle with long-range dependencies and global context modeling. Though atrous convolution expands receptive

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fields, it sacrifices fine-grained details. To address this, we design the ACPF²Net(Fig.3) encoder that fuses multi-scale features from an atrous convolution pyramid with low-level detail branches.

The encoder first extracts preliminary features using a ResNet backbone. To enhance context modeling—a known limitation of standard CNNs—these features are processed through an atrous convolution layer that expands the receptive field. Multi-scale features from the atrous pyramid are resized, concatenated, and fused via dimensionality-reduction convolution. While the enlarged receptive field captures richer context, it risks detail degradation. To compensate, early-stage residual features (preserving fine details) are integrated with the context-rich features through skip connections. This design enables simultaneous preservation of spatial details and contextual semantics without computational overhead.

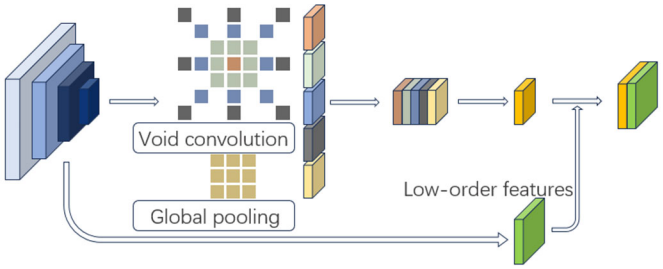


Fig 3 ACPF²Net model

As illustrated in Fig.4, the atrous convolution pyramid employs hierarchical feature fusion: (1) concatenated multi-scale features undergo low-pass filtering (LPF) via convolution kernels, then fused with upsampled global context from average pooling; (2) complementary high-pass filtered (HPF) features—implemented through inverted LPF kernels—are fused at matching resolutions. This dual-filter design suppresses cloud-induced low-frequency interference while enhancing critical high-frequency patterns (edges, contours, shadow textures). The hierarchical architecture reduces computational complexity compared to full-resolution processing.

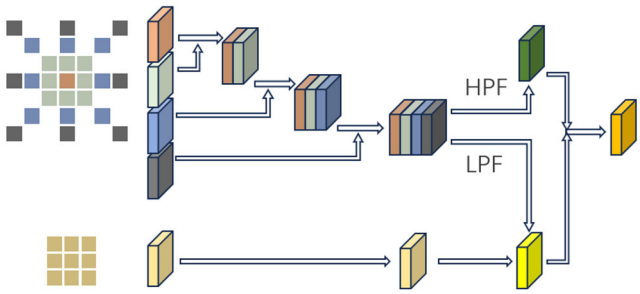


Fig 4 Feature Fusion Process in the ACPF²Net Model

III. Experiment

A. The experimental environment

In this paper, the whole algorithm is improved by using the Pytorch framework in Python3.9 environment. The operating system is Ubuntu 20.04, the CPU is Intel ® Core i7-10700F CPU @2.90GHz×16, the GPU is NVIDIA GeForce RTX 3070, and the acceleration environment is CUDA 11.7.

B. Datasets and evaluation indicators

The ISPRS Potsdam and Vaihingen datasets (German urban/rural benchmarks) provide reliable semantic annotations for remote sensing imagery, serving as standard benchmarks for domain-adaptive segmentation. Due to the absence of annotated maritime ship datasets, we evaluate our method using cloud-synthesized ISPRS variants and develop a dedicated CloudyShip dataset with realistic cloud cover to validate generalization capability. Our custom dataset comprises 3,100 satellite images with hierarchical semantic annotations: Primary task: Ship/background segmentation; Secondary task: Fine-grained classification(class distribution in Table 1)Annotation examples demonstrating cloud interference patterns are visualized in Figure 5.

Table.1 The number of samples for each class of self-made dataset

Class	aircraft carrier	warship	merchant vessel	submarine
Number	312	1762	2089	595



Fig 5 The annotation examples of self-made dataset

The primary metric, mean Intersection over Union (mIoU), is computed via the confusion matrix—a 2D array comparing predicted vs. actual class distributions. In this matrix:

Rows: Ground-truth classes | Columns: Predicted classes

Diagonal elements: Correct classifications

mIoU calculates the ratio of diagonal elements to all matrix elements, emphasizing classification errors and making it the gold standard for segmentation tasks. For instance-level evaluation (e.g., individual ships), Average Precision (AP) derived from precision-recall curves supplements mIoU analysis.

C. Results comparison

Domain Adaptation Benchmark Under Synthetic Cloud Conditions We conduct cross-domain segmentation experiments using cloud-synthesized ISPRS datasets: Potsdam (source domain, clear) → Vaihingen-C (target domain, synthetic clouds, Table 2). To focus on core maritime targets, we exclude ISPRS background classes and evaluate on five universal categories.

Evaluation Protocol: mIoU: Primary metric for semantic consistency; mAP: Pixel-wise average precision (no IoU

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thresholding); aAcc: Average class accuracy; F1-score: Harmonic mean of precision/recall

Tables 2 and 3 present ablation studies on domain-adaptive semantic segmentation under two cross-domain configurations:

Potsdam (source) → Cloud-Synthesized Vaihingen (target)

Vaihingen (source) → Cloud-Synthesized Potsdam (target)

Visualization: Figure 6: Table 2; Figure 7: Table 3

Stacked bars (blue=mIoU, orange=mAP, green=aAcc, red=F1-score) reveal relative strengths across metrics.

Table.2 Comparative experiment on P2V domain adaptation semantic segmentation under synthetic cloud conditions

Method	mIoU	mAP	aAcc	F1
Only Source (Deeplabv3)	31.51	44.12	55.08	49.43
Adapt-seg	44.97	52.38	63.82	60.32
ProDA	50.20	63.87	67.37	65.97
SAC	53.20	65.38	69.34	67.39
TriADA	56.24	71.48	70.72	69.26
DANN	58.02	60.27	69.59	79.36
ADANet	55.22	67.27	70.26	70.37
AEM-CurST-DA (ACPF²Net-1)	60.11	80.36	84.25	76.77

Table.3 Comparative experiment on V2P domain adaptation semantic segmentation under synthetic cloud conditions

Method	mIoU	mAP	aAcc	F1
Only Source (Deeplabv3)	30.74	42.45	53.47	48.38
Advent	46.66	59.23	68.37	64.92
DualGAN	50.67	62.65	66.09	64.23
DPL	54.47	66.73	69.91	67.38
ADANet	55.22	67.27	70.26	70.37
AEM-CurST-DA (ACPF²Net-1)	59.01	78.38	82.09	73.87

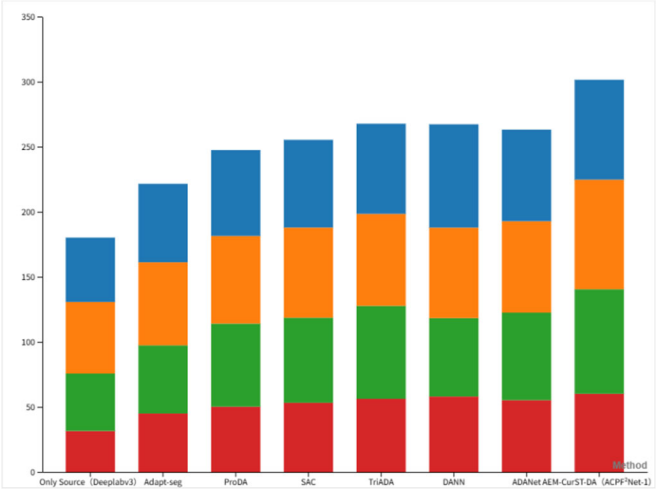


Fig 6 Visualization of experimental results of P2V domain adaptation semantic segmentation under synthetic cloud conditions

Analysis of Experimental Results: Directional Domain Shift: Domain adaptation difficulty varies with adaptation direction. For instance, Vaihingen→Potsdam adaptation exhibited a larger domain shift than the reverse. Method Superiority: Our framework (ACPF²Net-1 + AEM-CurST-DA) achieves state-of-the-art performance for cloudy optical satellite imagery:

mIoU: ~5% improvement over existing methods

mAP: 80.36% (Potsdam→Vaihingen) & 78.38% (Vaihingen→Potsdam)

D. Robustness Testing Under Extreme Conditions

To evaluate the robustness of the AEM-CurST-DA domain adaptation framework, multiple extreme weather scenarios were designed for domain adaptation experiments, including ablation studies under extreme dense fog and thick cloud conditions.

Extreme Dense Fog Scenario Experiment

Ablation experiments were conducted for the cross-domain semantic segmentation task of optical remote sensing images under extreme dense fog conditions.

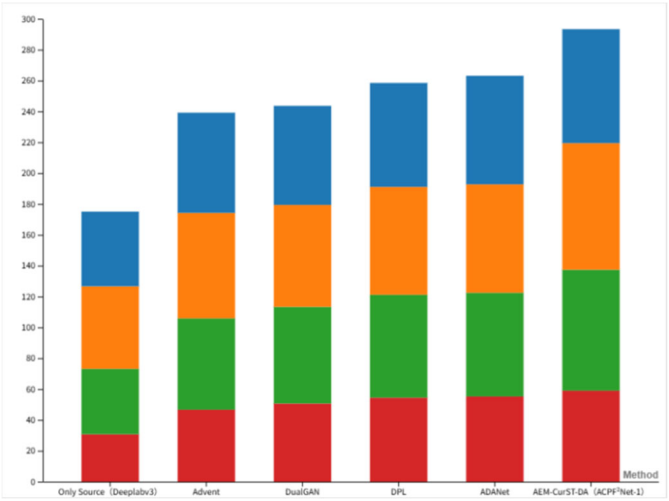


Fig 7 Visualization of experimental results of V2P domain adaptation semantic segmentation under synthetic cloud conditions

The extreme dense fog scenario is illustrated in Figure 8, and the corresponding experimental results are summarized in Table 4.



Fig 8 Dataset slices under extreme fog conditions

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Table.4 Ablation experiment of P2V domain adaptation to semantic segmentation under extreme fog conditions

Baseline	Framework	CurST	mIoU/%
ACPF²Net$_{-l}$	AEM-DA	w/o	48.50
ACPF²Net$_{-l}$	AEM-CurST-DA	with	49.67
AOD-net + deeplabv3	AEM-DA	w/o	47.90
AOD-net + deeplabv3	AEM-CurST-DA	with	48.89

Based on the experimental results in Table 4, introducing the curriculum-style self-training strategy under synthetic fog conditions improves the mIoU by 1.17%. However, the non-end-to-end training strategy combining a dehazing network with a semantic segmentation network resulted in a decrease in mIoU. This demonstrates that image distortion caused by the dehazing process negatively impacts semantic segmentation tasks, particularly in cross-domain semantic segmentation workflows.

Extreme Thick Cloud Scenario Experiment

To enhance the comprehensiveness of cloud and fog scenarios, further ablation experiments were conducted for the cross-domain semantic segmentation task of optical remote sensing images under synthetic thick cloud conditions. The extreme thick cloud scenario is illustrated in Figure 9, and the corresponding experimental results are summarized in Table 5.



Fig 9 Dataset slices under extreme cloud conditions

Table.5 Ablation experiment of P2V domain adaptation to semantic segmentation under extreme cloud conditions

Baseline	Framework	CurST	mIoU/%
ACPF²Net$_{-l}$	AEM-DA	w/o	48.63
ACPF²Net$_{-l}$	AEM-CurST-DA	with	49.88
Dehazeformer + deeplabv3	AEM-DA	w/o	50.12
Dehazeformer + deeplabv3	AEM-CurST-DA	with	48.48

Semantic segmentation under thick cloud conditions is more challenging compared to thin clouds or fog, as occlusion demands greater focus on contextual information to infer obscured regions. Results in Table 5 suggest using Transformer-based cloud removal networks with global attention mechanisms paired with semantic segmentation networks for thick cloud scenarios, while avoiding curriculum-style self-training strategies. Non-end-to-end networks may overestimate pseudo-label quality, degrading performance.

In robustness tests under extreme weather, the domain adaptation framework maintained reliable prediction accuracy (<1% decline), demonstrating the AEM-CurST-DA framework's stability and resilience in harsh meteorological conditions.

E. Experimental of self-made dataset

The custom dataset was constructed by selecting satellite remote sensing images from publicly available sources, with semantic segmentation annotations for ship classification (two-tier recognition tasks). Unoccluded images were artificially augmented with synthetic clouds/fog as the source domain, while the remaining images formed the target domain for domain adaptation experiments.

As shown in Table 6, the algorithm achieved partial success in L1-level semantic segmentation (mAP.50 >85%). Limitations include:Persistent domain shift, preventing full alignment with supervised training performance.Convolutional network constraints: While dilated convolutions and low-level feature fusion captured contextual patterns, insufficient contextual awareness in recovered details degraded performance under heavy occlusion.

L2-level tasks underperformed compared to L1, particularly for warships and commercial vessels. Key issues:Annotation ambiguity between commercial vessels and warships.Extreme intra-class size variations in commercial vessels (e.g., small fishing boats vs. cargo ships), complicating feature generalization.

Table.6 Experimental results of semantic segmentation of self-made satellite remote sensing ship dataset

Task level	class	Precision/%	Recall/%	AP.50/%
L1	ship	85.8	89.2	88.3
	aircraft carrier	93.3	84.3	85.6
	warship	68.6	80.5	73.5
L2	merchant vessel	55.8	42.1	45.1
	submarine	80.7	95.9	93.2
	ALL	74.6	75.7	74.3

IV. Conclusion

This paper first addresses the issue of insufficient annotation guidance for supervised learning in images under cloudy and foggy conditions by designing an end-to-end unsupervised domain adaptation semantic segmentation framework named AEM-CurST-DA. To tackle the challenge of feature extraction in remote sensing images under such conditions, an end-to-end feature extraction and fusion encoder model named ACPF²Net is developed. Furthermore, a series of comparative experiments and ablation studies are conducted to validate the effectiveness and robustness of the proposed algorithm.

The proposed algorithm achieves mIoU scores of 50.13% in the domain adaptation semantic segmentation experiment from the Potsdam dataset to the synthetic cloud-covered Vaihingen dataset, and 50.11% in the reverse adaptation from Vaihingen to synthetic cloud-covered Potsdam. When subjected to robustness testing under extreme dense fog and heavy cloud cover scenarios,

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the mIoU metric degradation remains controlled within approximately 1%, demonstrating its strong robustness. For cross-domain semantic segmentation on our custom marine vessel dataset, the network attains mAP values of 88.3% and 74.3% for L1-level and L2-level tasks respectively, highlighting the model's superior generalization capability across diverse maritime environments.

Collectively, the experimental results demonstrate that the feature fusion encoder ACPF²Net within the domain adaptation semantic segmentation framework AEM-CurST-DA, along with training strategies such as curriculum-style pseudo-labeling, significantly enhances the model's cross-domain segmentation capabilities. These synergistic improvements exhibit non-conflicting compatibility, confirming that the algorithmic design aligns with experimental hypotheses and effectively addresses semantic segmentation tasks for remote sensing imagery under cloudy and foggy conditions. Furthermore, evaluations on our custom benchmark dataset validate the algorithm's generalization prowess in real-world applications.

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